

What Open-Loop Hides and Closed-Loop Reveals: Weight Quantization, Generation Coherence, and On-Device Deployment of Autonomous-Driving Vision–Language–Action Models

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Abstract—Vision–language–action (VLA) models are an emerging approach to end-to-end autonomous driving, but their large language backbones must be compressed before they can run on in-vehicle hardware. Post-training quantization is the standard tool, yet its consequences for driving are unexamined—and the stakes differ fundamentally from the robot arms where VLA quantization has been studied: a driving policy shares the road with people, and a single corrupted plan is a collision. We present the first study of quantized driving VLAs, evaluated three ways: open-loop on nuScenes, closed-loop in the NeuroNCAP [5] photorealistic safety simulator, and on a real NVIDIA Jetson AGX Orin. The three views disagree in an instructive way. Open-loop success certifies 4-bit quantization as lossless—but we show this metric is dominated by ego-motion extrapolation and structurally cannot see quantization damage. Under closed-loop safety testing, the same quantized policy collapses: it stops producing coherent trajectories under the deployment-domain shift. The failure is governed by quantization scale granularity, not bit-width, and group-wise scales recover FP16-level behavior at negligible memory cost. On the Orin, the planner deploys comfortably, and both the accuracy and the open-loop metric’s blindness reproduce on the target hardware. Our conclusion for practitioners: quantized driving policies must be certified closed-loop and on-device—open-loop success is false assurance.

Index Terms—vision–language–action models, autonomous driving, post-training quantization, group-wise INT4, closed-loop evaluation, generation coherence, edge deployment, Jetson Orin

I. Introduction

On-board inference for end-to-end driving stacks motivates compressing the language backbones of driving VLAs. The prevailing evidence—near-lossless 4-bit weight quantization on open-loop nuScenes planning—suggests this is essentially free. We show this evidence is structurally unable to support a safety claim, that closed-loop evaluation tells a different and more actionable story, and that both conclusions reproduce on real edge hardware. We organize around five questions:

Q1 Can open-loop nuScenes L2 see quantization damage?

Q2 Does quantization affect closed-loop behavior, where the logged ego trajectory cannot be coasted on?

Q3 What property of the quantizer governs the effect, and what is the fix?

Q4 What is the memory/latency profile on an edge accelerator?

Q5 Do the accuracy conclusions hold when the model actually runs on that accelerator?

Contributions: a controlled demonstration of open-loop blindness ($19\times$ ego-ablation); the first closed-loop characterization of weight-quantization effects in a driving VLA; the identification of scale granularity (not bit-width) as the governing factor with a near-free fix; and end-to-end on-Orin systems and accuracy measurements, including a per-scene success breakdown.

II. Related Work and Positioning

Driving planners and VLAs. UniAD [2] introduced a planning-oriented, query-based perception stack (multi-camera ResNet-101, temporal BEVFormer [7], track/map/motion/ occupancy heads). VAD [3] vectorized the scene for efficiency. OpenDriveVLA [1] reuses UniAD track+map perception as a vision tower for a Qwen2.5 language/action head that emits trajectories as text, reaching state-of-the-art open-loop L2 (Table I). Open-loop critique. “Is Ego Status All You Need?” [4] showed nuScenes open-loop planning is dominated by ego state and proposed road-adherence metrics; we quantify the effect ($19\times$) specifically as it bears on quantization evaluation. Closed-loop neural simulation. NeuroNCAP [5] couples a NeuRAD [6] neural renderer to a safety-test protocol (frontal/side/stationary collision scenarios), enabling photorealistic closed-loop evaluation and revealing that open-loop-strong planners fail safety-critical closed-loop tests. LLM weight quantization. Round-to-nearest (RTN), GPTQ [9] (Hessian-weighted error minimization), and AWQ [8] (activation-aware scaling) differ in how they choose quantization grids; all rely on group-wise

(block) scales to bound weight-outlier damage, a practice motivated by the outlier phenomenon in LLMs [10], [11]. Our positioning. Prior quantization work targets language benchmarks (perplexity, MMLU); prior driving-eval work does not study quantization. We are, to our knowledge, the first to evaluate driving-VLA quantization closed-loop and on-device, and to show that the standard open-loop metric certifies a policy the closed-loop test rejects.

III. Method

A. Model and quantization scope

OpenDriveVLA-0.5B pairs Qwen2.5-0.5B (hidden 896, 24 layers) with UniAD stage-1 track+map perception. We apply weight-only post-training quantization to the 168 Linear layers of the Qwen backbone (357.8 M parameters), leaving perception, the scene/track/map projectors, token embeddings, and the output head in floating point, so any observed effect is attributable to the language head’s W_b precision. Weights are dequantized to the compute dtype for inference (fake-quant): the arithmetic path is unchanged, so accuracy effects are real while packed memory/latency gains are projected until a packed kernel is added (Sec. IV-D).

B. Quantization methodology and why granularity matters

Round-to-nearest (RTN). For a weight block $\mathbf{w}$ and b bits, symmetric RTN uses a single scale $s = \max_i |w_i| / (2^{b-1} - 1)$ and maps $\hat{w}_i = s \cdot \text{clip}(\text{round}(w_i/s), -2^{b-1}, 2^{b-1}-1)$; asymmetric RTN adds a zero-point over $[\text{min}, \text{max}]$. We use RTN deliberately: it is data-free and introduces no optimization of its own, so varying only the block over which s is shared isolates the effect of scale granularity.

Granularity. Per-output-channel shares one scale across an entire weight row (all 896/intermediate inputs); group-wise ($g=128$) uses an independent scale per contiguous block of g input weights. The extra storage is one FP16 scale per group, i.e. $16/g$ bits/weight (0.125 at $g=128$; Table V).

Why group-wise works (the outlier-scale mechanism). The shared scale is set by the block’s largest magnitude, $s = \max |w| / (2^{b-1} - 1)$. A single outlier weight inflates s and therefore coarsens the quantization grid for every small, signal-carrying weight that shares the scale. The effective per-weight rounding error is $\propto s$; at INT4 the grid has only 15 levels, so one outlier in a 896-wide row can multiply the error of the whole row. Group-wise scales confine each outlier to its own 128-weight block, bounding the damage. This is the same rationale behind LLM outlier handling [10], [11] and behind the group-wise default in GPTQ and AWQ; our contribution is to show it is decisive for closed-loop driving behavior, not merely for perplexity. Relation to prior methods (Table II): GPTQ additionally minimizes a Hessian-weighted reconstruction error via sequential updates; AWQ additionally rescales

salient channels using activation statistics. Both are complementary to—and layer on top of—the granularity we isolate; we use plain group-wise RTN to attribute the closed-loop recovery purely to granularity, and note GPTQ/AWQ would be expected to match or exceed it at equal bits.

C. Open-loop evaluation

We evaluate on the 162 nuScenes-mini $\cap$ validation samples under the ST-P3 and UniAD L2 protocols. The ego-ablation zeroes the ego-state vector and the 2s history trajectory in the planner prompt, leaving perception untouched, to test how much of L2 is ego-extrapolation.

D. Closed-loop harness

We built a native (Docker-free) three-process harness communicating over localhost HTTP: (i) the NeuroNCAP orchestrator (kinematic vehicle model, collision check, NCAP scoring); (ii) the NeuRAD renderer serving six photorealistic camera views at the ego’s driven pose each step; (iii) an OpenDriveVLA model node that reconstructs the UniAD perception input from rendered frames (normalization/padding, lidar-to-image projection, an 18-vector CAN-bus signal, ego-state prompt) and exposes an in-place re-quantization endpoint so techniques are swept without reloading. We verified the renderer produces photorealistic frames (per-camera intensity std 25–49 after installing tiny-cuda-nn; a missing-tcn fallback had previously rendered flat-grey blanks, and all pre-fix closed-loop numbers were discarded).

E. Metric: generation coherence

Our primary closed-loop metric is generation coherence: the fraction of frames on which the planner emits a well-formed (non-degenerate) trajectory, computed from the logged per-frame plans. A degenerate all-zero plan corresponds to malformed/empty model output that triggers a stay-in-place fallback. We treat collision/NCAP as a secondary, confounded proxy (Sec. IV-B).

F. On-device methodology (capture–replay)

To measure accuracy on the Orin without porting UniAD’s custom CUDA perception ops, we exploit that the fused planner input (inputs_embeds) is quantization-independent (perception and token embeddings are unquantized). We capture the fused inputs for each nuScenes sample from the A100 full pipeline, transfer them to the Orin, replay them through the decoder-only planner at each precision, and score the emitted trajectories with the same evaluator. This isolates device and quantization while using the real perception features and real weights.

IV. Results

A. Q1: Open-loop L2 is blind

Weight quantization is lossless to INT4; the INT3/INT2 degradation is generation collapse (malformed text), not graded planning. The ego-ablation shows the metric is

TABLE I

Open-loop nuScenes planning context (ST-P3 L2, meters; lower better). Baseline numbers as reported by the respective papers; OpenDriveVLA-0.5B is our study model. Our contribution is not a new planner but the observation that this metric cannot separate FP16 from quantized policies.

Method	1s	2s	3s	Avg.	Coll. avg (%)
UniAD [2]	0.44	0.67	0.96	0.69	0.12
VAD [3]	0.17	0.34	0.60	0.37	0.14
OpenDriveVLA-7B [1]	0.15	0.31	0.55	0.33	0.10
OpenDriveVLA-3B [1]	0.14	0.30	0.55	0.33	0.10
OpenDriveVLA-0.5B (ours)	0.15	0.32	0.57	0.35	0.09

TABLE II

Weight-only PTQ methods. Our study isolates granularity with RTN; GPTQ/AWQ add orthogonal error-minimization on top of group-wise scales.

Method	data-free	grid choice	granularity	extra cost	studied here
RTN per-channel	yes	max $ w $	per-row	—	baseline (collapses)
RTN group-128	yes	max $ w $ /block	per-128	+0.125 b/w	fix
GPTQ [9]	no (calib.)	Hessian-weighted	group-wise	calib. pass	related
AWQ [8]	no (calib.)	activation-scaled	group-wise	calib. pass	related

TABLE III

Open-loop nuScenes-mini ($n=162$), ST-P3 L2 (m).

Setting	L2 (m)	well-formed
FP16	0.33	100%
INT8 (per-channel)	0.33	100%
INT4 (per-channel)	0.33	100%
INT3	1.91	92%
INT2	6.45	0%
FP16, ego-state+history zeroed	6.38 (19 $\times$)	100%

$\approx$ ego-extrapolation: with ego state removed the plan collapses 19 $\times$ while outputs stay well-formed. Open-loop L2 thus cannot reflect perception/planning, nor quantization damage to them—the same phenomenon that makes 0.5B/3B/7B report near-identical L2 (Table I).

B. Q2–Q3: Closed-loop reveals collapse; granularity is the fix

Across all 16 scenario instances, per-channel INT4 collapses generation ($\leq 15\%$ well-formed on 15/16, mean 3.7%) while group-wise INT4 recovers the bulk of FP16 coherence (mean 73.5% vs 76.0%, $\geq 85\%$ of FP16 on 13/16). The same per-channel INT4 is lossless in-distribution (Table III), so the failure is a quantization $\times$ domain-shift interaction, governed by the granularity mechanism of Sec. III-B. Group-wise only partially recovers on 3 scenarios (0099, 0101, 0796); we do not claim exact equivalence.

a) Collision/NCAP is a confounded proxy.: On several scenes (e.g. 0796, 0278-stationary) every configuration including FP16 collides 10/10: the base policy cannot clear those under our rendering, so collisions cannot isolate quantization there even though coherence still separates the configs. We therefore lead with coherence.

C. Q4: Memory and edge profile

The robust recipe (group-wise INT4) is ≈ 0.26 GB; the whole node fits in 3–5 GB, so the planner is trivial for a Jetson AGX Orin. The obstacle is perception throughput: UniAD’s six-camera ResNet-101 + deformable-attention BEVFormer runs FP32 with custom CUDA kernels, far from real-time on Orin without a TensorRT-INT8 port. Edge effort should target perception; the quantized planner is a solved sub-problem.

D. Q5: On-device measurement (Jetson AGX Orin 64 GB)

We ran the planner on a real AGX Orin 64 GB (JetPack 6 / L4T R36.4.7, CUDA cap 8.7, PyTorch 2.8) with the actual OpenDriveVLA-0.5B decoder weights.

a) On-device accuracy reproduces the A100 and the published result.: Replaying real inputs (Sec. III-F) on the Orin over all 162 mini-val samples yields FP16 ST-P3 L2 0.146/0.317/0.571 m (1/2/3s; avg 0.345m)—matching OpenDriveVLA-0.5B’s reported 0.35m (Table I)—with group-INT4 0.151/0.327/0.583 and per-channel-INT4 0.156/0.362/0.652. The Orin reproduces A100 trajectories to ≈ 1.1 cm mean (FP16) / 1.5 cm (group-INT4). Crucially, the per-sample W4 plans already differ from FP16 by 14–24 cm while aggregate L2 moves

TABLE IV
NeuroNCAP closed-loop generation coherence (well-formed predicted-frame rate, %), real-camera NeuRAD rendering, 14 scenes / 16 scenario instances, 10 paired seeds each.

scene / scenario	FP16	INT4 per-channel	INT4 group-128
0099 stationary	60.6	0.0	30.0
0101 stationary	75.0	14.3	46.7
0103 frontal	74.3	0.0	72.9
0106 frontal	66.9	0.0	78.7
0108 side	85.7	0.0	91.6
0108 stationary	72.4	0.0	74.7
0110 frontal	70.7	0.0	74.3
0278 side	75.3	0.0	67.9
0278 stationary	79.7	0.0	83.5
0331 stationary	83.2	5.0	97.5
0346 frontal	83.1	15.6	86.9
0783 stationary	72.4	0.0	73.1
0796 stationary	72.4	8.6	57.9
0921 side	90.0	14.2	90.4
0923 frontal	74.6	0.8	72.1
0966 stationary	79.4	0.0	78.3
mean	76.0	3.7	73.5

TABLE V
Backbone weight footprint (357.8 M params); effective bits include per-group FP16 scales.

Precision	eff. bits/weight	backbone weights
FP16	16.0	≈ 1.0 GB
INT8 per-channel	8.06	≈ 0.50 GB
INT4 per-channel	4.06	≈ 0.25 GB
INT4 group-128	4.125	≈ 0.26 GB

TABLE VI
Measured on AGX Orin. Planner systems profile (1024-token context $\rightarrow$ 60 decoded tokens, eager attention, default clocks, median of 30). Latency is identical across precisions (fake-quant); the deployable win is the static packed footprint.

Config	median latency	throughput	runtime peak	packed weights
FP16	4.02 s	14.9 tok/s	2.88 GB	0.716 GB
INT8 group-128	4.01 s	15.0 tok/s	2.88 GB	0.363 GB
INT4 group-128	4.02 s	14.9 tok/s	2.88 GB	0.185 GB
INT4 per-channel	4.02 s	14.9 tok/s	2.88 GB	0.179 GB

< 3 cm: open-loop L2 averages the perturbation away on the deployment hardware itself.

Overall $\sim 80\%$ of plans land within 1 m of the human path (88% within 1.5 m), but this ranges 58.5% (dynamic parking-lot maneuvering) to 97.6% (a stationary “wait at intersection” clip where “predict stopped” is trivially near-exact—an instance of the ego-extrapolation effect of Q1). Open-loop success cannot separate the precisions even per scene, which is precisely why closed-loop coherence (Table IV) is the appropriate metric.

V. Discussion

The study composes into one message: open-loop L2 hides quantization damage because it is ego-extrapolation; closed-loop reveals it because perception and planning are on the critical path; the revealed damage is a granularity artifact with a near-free fix; and both the blindness and the fix reproduce on real Orin hardware. Practitioners should

(i) never certify a quantized driving policy on open-loop L2, and (ii) default to group-wise INT4 for the planner.

VI. Threats to Validity

Scene coverage: closed-loop spans all 14 released NeuroNCAP scenes (16 scenario instances), not all of nuScenes. Renderer domain gap: rendered inputs are OOD to real nuScenes; on several scenes even FP16 fails the collision test (10/10), so absolute NCAP numbers are not in-distribution-comparable and our claims are relative. Partial recovery: group-wise ties/ beats FP16 on 13/16 but partially recovers on 3 (0099, 0101, 0796). Metric proxy: collision/ NCAP is confounded; we lead with coherence. Fake-quant: the static footprint is measured on-device, but a packed-INT4 speedup needs a custom kernel (Marlin/AWQ-style) + TensorRT and is not yet realized—hence flat on-Orin latency. On-Orin accuracy scope: planner-only replay (perception not ported to sm_87), fp16 replay, mini-val. AWQ/INT3/INT2: characterized

TABLE VII

Measured on AGX Orin. Per-scene open-loop success: % of predicted plans within a distance of the human (GT) trajectory (mean L2 over 3s), all 162 mini-val samples, 0 malformed. Group-INT4 tracks FP16 per scene; success varies far more by scene difficulty than by precision.

scene (character)	N	mean L2	≤ 1.0 m	≤ 1.5 m	≤ 2.0 m
0553 (wait at intersection)	41	0.00	97.6%	97.6%	97.6%
0796 (busy street)	40	0.30	95.0%	95.0%	95.0%
0103 (peds, turning car)	40	0.88	67.5%	85.0%	85.0%
0916 (parking-lot maneuver)	41	1.10	58.5%	75.6%	85.4%
ALL (FP16)	162	0.57	79.6%	88.3%	90.7%
ALL (INT4 group-128)	162	0.58	78.4%	88.3%	90.1%
ALL (INT4 per-channel)	162	0.65	80.9%	87.7%	89.5%

only in an earlier single-scene, pre-renderer-fix setup and not re-reported here.

[11] G. Xiao et al. SmoothQuant: Accurate and Efficient Post-Training Quantization for Large Language Models. ICML 2023. arXiv:2211.10438.

VII. Reproducibility

The public release (github.com/Omega1029/DrivingVLA) includes: the open-loop eval and quantization sweeps with the ego-ablation toggle; the three-process closed-loop harness and in-place re-quantization service; the 4-GPU-lane 14-scene benchmark runner; the on-device suite (`orin/bench_planner_orin.py`, `replay_accuracy.py`, `subset_l2.py`, `scene_success.py`) and result JSONs; and the closed-loop video renderers. All numbers here are produced by those scripts; estimates are marked.

VIII. Conclusion

For quantized autonomous-driving VLAs, open-loop L2 is blind ($19\times$ ego-ablation) and closed-loop is necessary: there, per-channel INT4 collapses generation under deployment-domain shift while group-wise INT4 (≈ 0.1 extra bits/weight) matches FP16. On a real AGX Orin the planner fits in 2.88 GB, packs $3.9\times$ at group-INT4, and reproduces both the published accuracy and the metric’s blindness. Use closed-loop coherence as the metric and group-wise INT4 as the recipe.

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